

Inventory Management

Elements of Inventory Management

Role of Inventory

- Inventory is a stock of items kept on hand used to meet customer demand.
- A level of inventory is maintained that will meet anticipated demand.
- If demand not known with certainty, safety (buffer) stocks are kept on hand.
- Additional stocks are sometimes built up to meet seasonal or cyclical demand.

Elements of Inventory Management

Role of Inventory

- Large amounts of inventory sometimes purchased to take advantage of discounts.
- In-process inventories maintained to provide independence between operations.
- Raw materials inventory kept to avoid delays in case of supplier problems.
- Stock of finished parts kept to meet customer demand in event of work stoppage.

Elements of Inventory Management

Demand

- Inventory exists to meet the demand of customers.
- Customers can be external (purchasers of products) or internal (workers using material).
- Management needs accurate forecast of demand.
- Items that are used internally to produce a final product are referred to as *dependent demand items*.
- Items that are final products demanded by an external customer are *independent demand items*.

Elements of Inventory Management

Inventory Costs

- *Carrying costs* - Costs of holding items in storage.
 - ◆ Vary with level of inventory and sometimes with length of time held.
 - ◆ Include facility operating costs, record keeping, interest, etc.
 - ◆ Assigned on a per unit basis per time period, or as percentage of average inventory value.

Elements of Inventory Management

Inventory Costs

- *Ordering costs* - costs of replenishing stock of inventory.
 - ◆ Expressed as amount per order, independent of order size.
 - ◆ Vary with the number of orders made.
 - ◆ Include purchase orders, handling, inspection, etc.

Elements of Inventory Management

Inventory Costs

- *Shortage, or stockout costs* - Costs associated with insufficient inventory.
 - ◆ Result in permanent loss of sales and profits for items not on hand.
 - ◆ Sometimes penalties involved; if customer is internal, work delays could result.

Inventory Control Systems

- ☀ An inventory control system controls the level of inventory by determining *how much* (replenishment level) and *when* to order.
- ☀ Two basic types of systems -*continuous (fixed-order quantity)* and *periodic (fixed-time)*.
- ☀ In a continuous system, an order is placed for the same constant amount when inventory decreases to a specified level.
- ☀ In a periodic system, an order is placed for a variable amount after a specified period of time.

Inventory Control Systems

Continuous Inventory System

- ☀ A continual record of inventory level is maintained.
- ☀ Whenever inventory decreases to a predetermined level, the *reorder point*, an order is placed for a *fixed amount* to replenish the stock.
- ☀ The fixed amount is termed the *economic order quantity*, whose magnitude is set at a level that minimizes the total inventory carrying, ordering, and shortage costs.
- ☀ Because of continual monitoring, management is always aware of status of inventory level and critical parts, but system is relatively expensive to maintain.

Inventory Control Systems

Periodic Inventory System

- ☀ Inventory on hand is counted at specific time intervals and an order placed that brings inventory up to a specified level.
- ☀ Inventory not monitored between counts and system is therefore less costly to track and keep account of.
- ☀ Results in less direct control by management and thus generally higher levels of inventory to guard against stockouts.
- ☀ System requires a new order quantity each time an order is placed.
- ☀ Used in smaller retail stores, drugstores, grocery stores and offices.

Economic Order Quantity Models

- ☀ *Economic order quantity*, or economic lot size, is the quantity ordered when inventory decreases to the *reorder point*.
- ☀ Amount is determined using the economic order quantity (EOQ) model.
- ☀ Purpose of the EOQ model is to determine the optimal order size that will minimize total inventory costs.
- ☀ Three model versions to be discussed:
 - ◆ Basic EOQ model
 - ◆ EOQ model without instantaneous receipt
 - ◆ EOQ model with shortages

Economic Order Quantity Models

Basic EOQ Model

- ✦ A formula for determining the optimal order size that minimizes the sum of carrying costs and ordering costs.
- ✦ Simplifying assumptions and restrictions:
 - ◆ Demand is known with certainty and is relatively constant over time.
 - ◆ No shortages are allowed.
 - ◆ Lead time for the receipt of orders is constant.
 - ◆ The order quantity is received all at once and instantaneously.

Economic Order Quantity Models

Basic EOQ Model

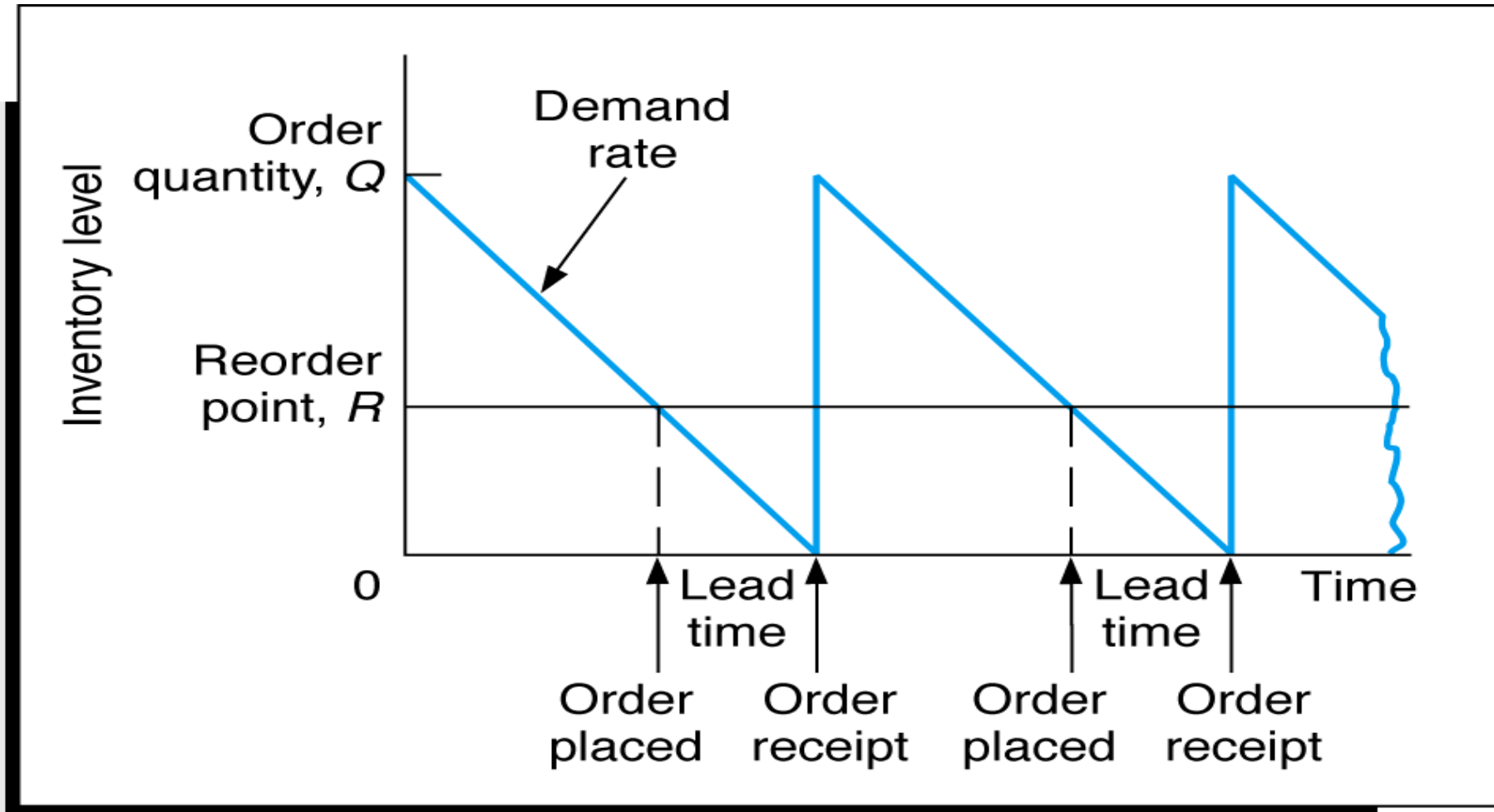


Figure The Inventory Order Cycle

Basic EOQ Model

Carrying Cost

- ☀ Carrying cost usually expressed on a per unit basis of time.
- ☀ Annual carrying cost equals carrying cost per unit per year times average inventory level:
 - ◆ Carrying cost per unit per year = C_c
 - ◆ Average inventory = $Q/2$
 - ◆ Annual carrying cost = $C_c Q/2$.

Basic EOQ Model

Carrying Cost

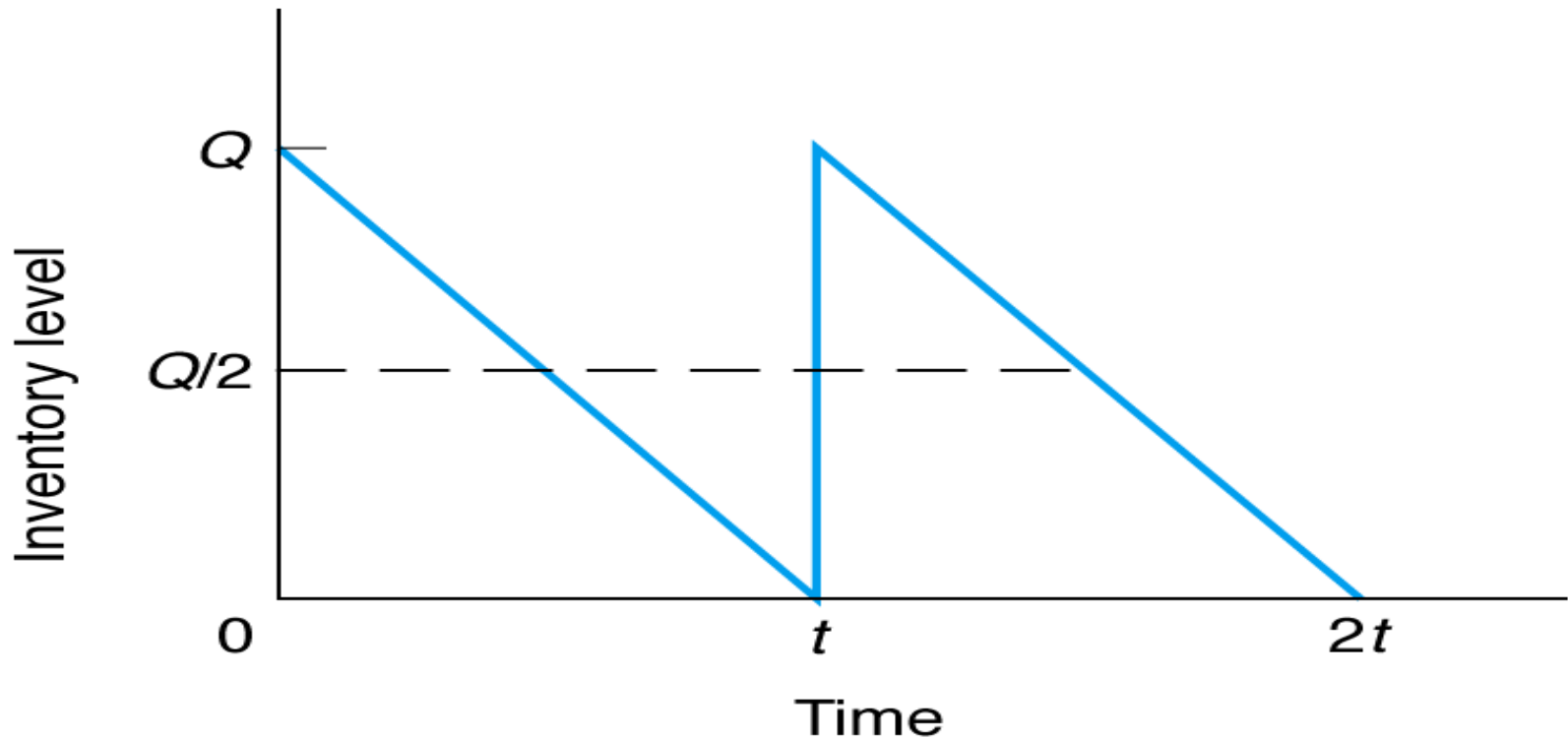


Figure: Average Inventory

Basic EOQ Model

Ordering Cost

- ✱ Total annual ordering cost equals cost per order (C_o) times number of orders per year.
- ✱ Number of orders per year, with known and constant demand, D , is D/Q , where Q is the order size:
 - ✱ Annual ordering cost = $C_o D/Q$
 - ✱ Only variable is Q , C_o and D are constant parameters.

Basic EOQ Model

Total Inventory Cost

- ✦ Total annual inventory cost is sum of ordering and carrying cost:

$$TC = C_o \frac{D}{Q} + C_c \frac{Q}{2}$$

TC = Total Inventory Cost

C_o = Ordering Cost

C_c = Inventory Carrying Cost

Q = Order Quantity

D = Demand

Basic EOQ Model

Total Inventory Cost

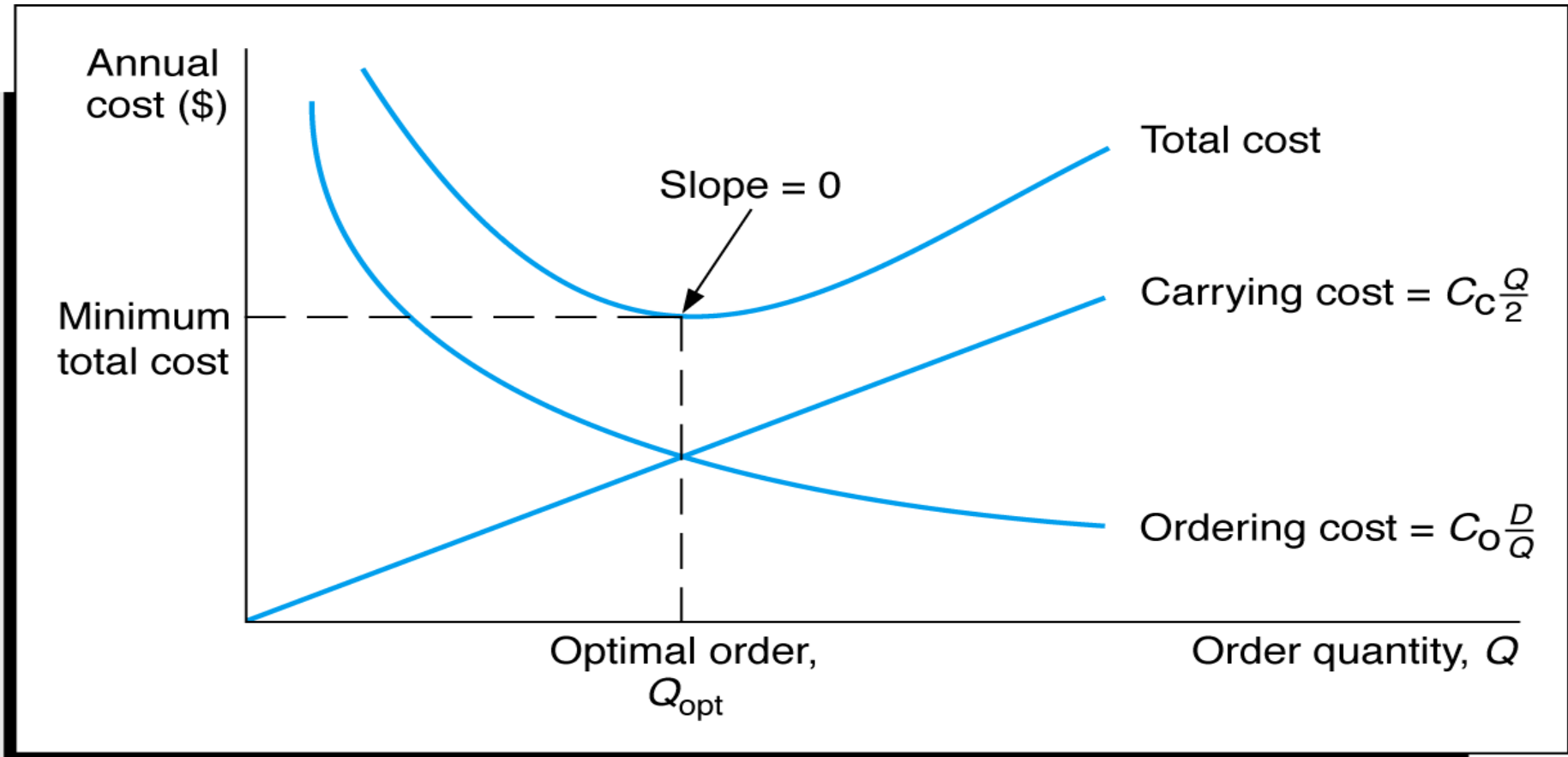


Figure: The EOQ Cost Model